BRD — Offline Bangla Dubbing Video Player

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A cross-platform desktop video player that dubs any spoken language into Bangladeshi Bangla voice, offline, with a distinct voice per speaker.

Version 1.1. Updated after the spike and the rework plan: fallbacks that would otherwise need a rework loop are now built into the requirements. See Decisions and change log at the end.

Business problem

Bangla speakers who do not understand the original language cannot follow most foreign films, lectures and shows. Subtitles demand constant reading, which excludes children, elderly viewers, people with low literacy or poor eyesight, and anyone multitasking.

Professional Bangla dubbing is expensive, slow, and only exists for a small share of content. Existing AI dubbing services are online-only, paid per minute, and send private videos to third-party servers. Few of them offer Bangladeshi Bangla at all.

The gap: no offline player turns any video into spoken Bangladeshi Bangla on the user's own computer.

Objectives and success metrics

The proposed targets below are starting values to confirm during gap analysis.

Objective	Success metric (proposed target)
Understandable Bangla dubbing	80%+ of test viewers follow the story without subtitles
Correct speaker voices	Speaker labels correct in 90%+ of lines with 2–4 speakers
Fully offline	Zero network calls after first-run model download
Fast enough to use	Pre-process: under 15 min per 1 hour of video on an 8 GB NVIDIA GPU; live mode starts within 60 s
Stays in sync	Dubbed line starts within 0.5 s of the original line

Objective	Success metric (proposed target)
Cross-platform	Installers for Windows, macOS and Linux pass the same test set

Scope

In scope (version 1)

- Desktop player built with PyQt for Windows, macOS and Linux
- Common video formats (MP4, MKV, AVI, MOV, WebM) through ffmpeg
- Automatic source-language detection and speech-to-text
- Translation into Bangla
- Bangla text-to-speech with a distinct voice per detected speaker
- Original music and sound effects kept under the dubbed voices
- Pre-process mode and buffered live mode
- Export of the dubbed audio track or a dubbed video file
- Optional Bangla subtitle display from the same translation

Out of scope (version 1)

- Lip-sync or any change to the video picture
- Cloning each original actor's voice into Bangla
- Streaming sites, YouTube URLs or live TV input
- Translation into languages other than Bangla
- Mobile apps and cloud processing

Stakeholders and target users

Stakeholder	Role / interest
Product owner	Owns the requirements, approves UAT and release
Development team	Builds the PyQt app and the AI pipeline
QA team	Tests dubbing quality, sync and each platform

Stakeholder	Role / interest
UAT testers	Native Bangladeshi Bangla speakers who judge naturalness
End users: home viewers	Watch foreign films and shows in Bangla, including children and elderly family members
End users: students	Follow foreign-language lectures and tutorials
End users: low-connectivity areas	Need everything to work without internet

Names for each internal role are still to be assigned.

Business requirements

Priority uses MoSCoW: Must, Should, Could.

ID	Requirement	Priority
BR-01	The user opens a local video file and plays it with standard controls (play, pause, seek, volume, fullscreen).	Must
BR-02	The system detects the spoken language automatically; the user can override it. Version 1 guarantees English and Hindi; other languages are supported through the English pivot on a best-effort basis.	Must
BR-03	The system replaces the original speech with spoken Bangla, not subtitles.	Must
BR-04a	The Bangla text uses Bangladeshi vocabulary (translation plus a Bangladeshi glossary).	Must
BR-04b	The Bangla voice has a Bangladeshi accent, via TTS fine-tuned on the SUBAK.KO corpus. Version 1 may ship with preset voices.	Should
BR-05	Each detected speaker gets a different, consistent Bangla voice for the whole video. Guaranteed for up to 4 speakers; up to 6 supported.	Must

ID	Requirement	Priority
BR-06	The system keeps original music and sound effects under the dubbed voices. If separation is poor, it falls back to lowering the original audio under the dub.	Should
BR-07	All processing runs on the user's computer with no internet after the one-time model download.	Must
BR-08	Pre-process mode dubs the whole video before playback, synced.	Must
BR-09	Buffered live mode starts playback within 60 s and keeps ahead of playback, on recommended hardware only. Not a release blocker for version 1.	Should
BR-10	The user can choose or change the voice assigned to each speaker.	Should
BR-11	The user can switch between original and Bangla audio during playback.	Should
BR-12	The user can export the dubbed audio or a dubbed video file.	Should
BR-13	The user can show Bangla subtitles from the same translation.	Should
BR-14	Processed videos are cached so replaying needs no reprocessing.	Could
BR-15	The user can review and edit the Bangla text of a line, then regenerate its voice.	Should
BR-16	The app interface is available in Bangla and English.	Could
BR-17	The user can rename, merge or split detected speakers and re-dub the affected lines.	Should

Non-functional requirements

ID	Area	Requirement
NFR-01	Platforms	Windows 10/11, macOS 12+ (Apple Silicon first), Ubuntu 22.04+
NFR-02	Hardware	Recommended: NVIDIA GPU with 8 GB+ VRAM or Apple Silicon with 16 GB RAM. Minimum: CPU only, pre-process mode only; live mode is disabled automatically
NFR-03	Privacy	No video, audio or text leaves the device
NFR-04	Storage	Models take about 5–10 GB, downloaded on first launch with a progress bar and resume
NFR-05	Sync	95% of dubbed lines start within 0.5 s of the original; no line more than 1.0 s late
NFR-06	Reliability	A failed line falls back to the original audio instead of stopping playback
NFR-07	Usability	A first-time user dubs a video in 3 clicks or fewer
NFR-08	Licensing	Every model and library permits commercial use; NLLB-200 and MMS-TTS are excluded
NFR-09	Packaging	One installer per platform (PyInstaller or Nuitka)
NFR-10	Speed	Pre-process mode under 15 min per 1 hour of video on recommended hardware

Assumptions, constraints and risks

Assumptions

- Users accept a one-time model download before going fully offline.
- "Instant" means a short buffer (30–60 s) or a pre-process step, not zero delay.
- Distinct preset Bangla voices per speaker are acceptable for version 1; a Bangladeshi-accent voice follows in phase 2.
- The product is commercial, so only commercially licensed models are used.

Constraints

- Built in Python with PyQt.
- Uses open-source models only, to stay offline and free of per-minute fees.
- Model stack: faster-whisper, pyannote 3.1, Demucs, IndicTrans2, Indic Parler-TTS.
- At most 2 rework loops per failed spike experiment before a not-feasible decision.

Risks

Risk	Impact	Mitigation (built into v1.1)
Open Bengali voices sound Kolkata, not Bangladeshi	Medium	BR-04 split: vocabulary is Must, accent is Should; fine-tune on SUBAK.KO (CC-BY-4.0) in phase 2
Diarization errors with many or overlapping speakers	Medium	BR-05 guaranteed for 4 speakers; BR-17 lets users fix speakers
Translation sounds literal or unnatural	Medium	BR-15 line editor raised to Should; local LLM as alternative translator
Bangla lines too long to stay in sync	Medium	NFR-05 allows 5% of lines up to 1.0 s; line editor shortens text
Live mode too slow on mid-range PCs	Low	BR-09 is not a release blocker; auto-disabled below recommended hardware
Poor background separation on dense music	Low	BR-06 falls back to lowering the original audio
Model licences block commercial release	Low	Resolved: commercial-safe stack chosen (NFR-08)
Large app and model size	Low	Download models on first run; offer smaller models

Open questions for gap analysis

- [x] ~~Is the product commercial, free, or internal? **Decided: commercial** (NFR-08).~~
- [x] ~~Which source languages matter most first? **Decided: English and Hindi guaranteed**, others best-effort (BR-02).~~

- [ ] What is the minimum hardware the target users actually own?
- [x] ~~Is a Bangladeshi voice dataset available? **Yes: SUBAK.KO**, 241 h, CC-BY-4.0 (BR-04b).~~
- [x] ~~Are a 30–60 s live buffer and preset voices acceptable? **Decided: yes for version 1.**~~
- [ ] What are the budget and target release date?
- [ ] Who are the product owner, developers, QA and UAT testers?

Decisions and change log

Decision: apply the rework plan's fallbacks now, before the spike runs. The likely failures (diarization, sync, translation, voice accent, live speed) then become reduced scope instead of rework loops back to step 1. Only an E1 failure (no working offline chain) still forces rework.

Fallback release: if the spike still fails after 2 rework loops, version 1 can ship as single-speaker dubbing plus Bangla subtitles, with the product owner's approval.

Version	Date	Change	Reason
1.1	2026-10-29	BR-02: English and Hindi guaranteed, others best-effort	Rework plan, E3
1.1	2026-10-29	BR-04 split into 04a vocabulary (Must) and 04b accent (Should)	Rework plan, E4; SUBAK.KO found in spike
1.1	2026-10-29	BR-05 guaranteed for 4 speakers; new BR-17 speaker fixing (Should)	Rework plan, E5
1.1	2026-10-29	BR-06 Must → Should, with audio-lowering fallback	Rework plan, E8
1.1	2026-10-29	BR-09 not a release blocker; auto-disabled on minimum hardware	Rework plan, E6
1.1	2026-10-29	BR-13 Could → Should; BR-15 Could → Should	Supports the fallback release and E2 / E7 fixes
1.1	2026-10-29	NFR-05: 95% within 0.5 s, none over 1.0 s	Rework plan, E7
1.1	2026-10-29	NFR-08: commercial licences only; NFR-10 speed added	Spike licence check

Version	Date	Change	Reason
1.0	2026-10-29	First version	—